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To cite this article: Shaohua Yan, Haiyang Zhou & Qing H. Qin (2019) Microstructure versus size: nano/microscale deformation of solute-strengthening Al alloys via pillar compression tests, Materials Research Letters, 7:2, 53-59, DOI: [10.1080/21663831.2018.1553802](https://doi.org/10.1080/21663831.2018.1553802)

To link to this article: <https://doi.org/10.1080/21663831.2018.1553802>



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ORIGINAL REPORT



Microstructure versus size: nano/microscale deformation of solute-strengthening Al alloys via pillar compression tests

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ABSTRACT

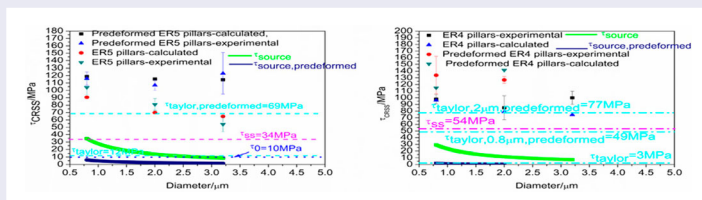
Effect of solutes, crystal orientation, sample's size, and dislocation density on the nano/microscale deformation of solute-strengthening Al alloys has been investigated via pillar compression tests. The Mg-solute-strengthening pillar with dislocation density of 10^{12} m^{-2} shows size- and orientation-dependent strength, while the strength of Si-Mg-solute-strengthening pillar with dislocation density of 10^{13} m^{-2} is size- and orientation- independent. With increasing the dislocation density, the strength of pillars increased and become size-independent.

ARTICLE HISTORY

Received 7 October 2018

KEYWORDS

Solutes strengthening; dislocation; compression test; mechanical properties



IMPACT STATEMENT

Links between the microstructure, sample's size and nano/microscale plasticity of solute-strengthening Al alloys were made via pillar compression testing. Deformation mechanism at nano/microscale was quantitatively explored.

1. Introduction

Uniaxial pillar compression has been the most popular testing methodology for detecting the plasticity of single metallic crystal at nano/microscales since 2004 [1,2]. Unlike the flow strength of the bulk, which is controlled by the Taylor hardening, such nano/microscale plastic deformation features with intermittent strain bursts and size-dependent strength [3–5]. Since dislocation is the only carrier of plastic deformation at such scales, several dislocation-based models have been proposed to explain the observed phenomenon [6–11], among which the two best known are the dislocation-starvation model and the single-ended dislocation source model. At nanoscale, dislocations in the pillar have the greater likelihood to escape from the sample's free surface than to interact with each other, thus resulting in a dislocation-starvation state with high flow stress [9]. At microscale, a double-ended

dislocation source (the Frank-Read source) in the pillar would become single-ended dislocation source due to the operation with the free surface, and the strength of the pillar is inverse to the maximum dislocation source (ϵR , radius of the pillar) [8]. A smaller pillar has shorter dislocation source, and thus the flow stress is higher.

Although extensive research has been conducted at nano/microscales, most works focus on the single crystals of pure metal, such as Ni [1], Cu [12] and Al [13]. However, in-service engineering materials are complex-microstructure alloys with solid solution, precipitates, and grain boundaries. Such internal obstacles, which can impede the dislocation motion under external loading, coupled with the size of the external sample, could alter the deformation behaviour of nano/microscale samples [14,15]. But a good understanding of the influence of the interplay between the internal microstructure and

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Supplemental data for this article can be accessed here. <https://doi.org/10.1080/21663831.2018.1553802>

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the size of the external sample on the strength is still lacking.

In this study, the microstructure of the pillar is modified with solute strengthening, which is a broadly used material-strengthening method in engineering materials (e.g. 5xxx aluminium alloys, fusion zone of welded Al alloys), and the dislocation density is modified by pre-deformation of the pillar. The effects of crystal orientation, solute elements and dislocation density on the size-dependent strength are studied.

2. Materials and methods

The materials used in this study are the fusion zone (FZ) of hybrid laser welded AA6061 with two filling materials (ER5356 and ER4043). More detail of the welding conditions can be found in Ref. [16].

The FZ was chosen as the studied material for two reasons: (1) the filling materials could produce different solute solutions and dislocation densities in the FZ, so that the effect of solutes and dislocation density on the mechanical properties of the nano/micropillar could be studied; (2) the plasticity of the nano/microscale pillar could be used to predict the macroscale properties [17]. The difference of FZs from these two filling materials can be found in Ref. [16].

The microstructure of two samples in the FZ are shown in Figure S1 in Supplementary materials. According to the EBSD results, as shown in Figure S2 in Supplementary materials, four orientations ($[10\bar{1}]$, $[111]$, $[31\bar{2}]$, and $[\bar{5}32]$) were selected for the sample in FZ of the ER4 joint, whereas two orientations ($[101]$ and $[111]$) were selected for the sample in FZ of the ER5 joint. The summary of crystallographic data for selected grains can be found in Table S1 in Supplementary materials. Pillars with diameter ranging from 400 nm to 3.3 μm were machined using FIB by FEI Helios 600Nanolab. The ratio of the pillar's height to diameter is kept between 3:1 and 4:1 to guarantee the pillar free of substrates constrain and not to be buckling during compression. For simplicity, we denote the pillars from the FZ of ER4 and ER5 joint as ER4 and ER5 pillars, respectively. The machined pillars were compressed by a nanoindenter (TriboIndenter TI900, Hysitron) with a flat-end tip (9 μm diameter). The thermal drift was controlled below 0.05 nm/s, and the strain rate is around $1 \times 10^{-3} \text{ s}^{-1}$. For each condition, three pillars were compressed. The dislocation density of the deformed pillar was examined by transmission electron microscope (TEM). The TEM experiment was conducted with JEOL 2100F microscope at the acceleration voltage of 200 kV.

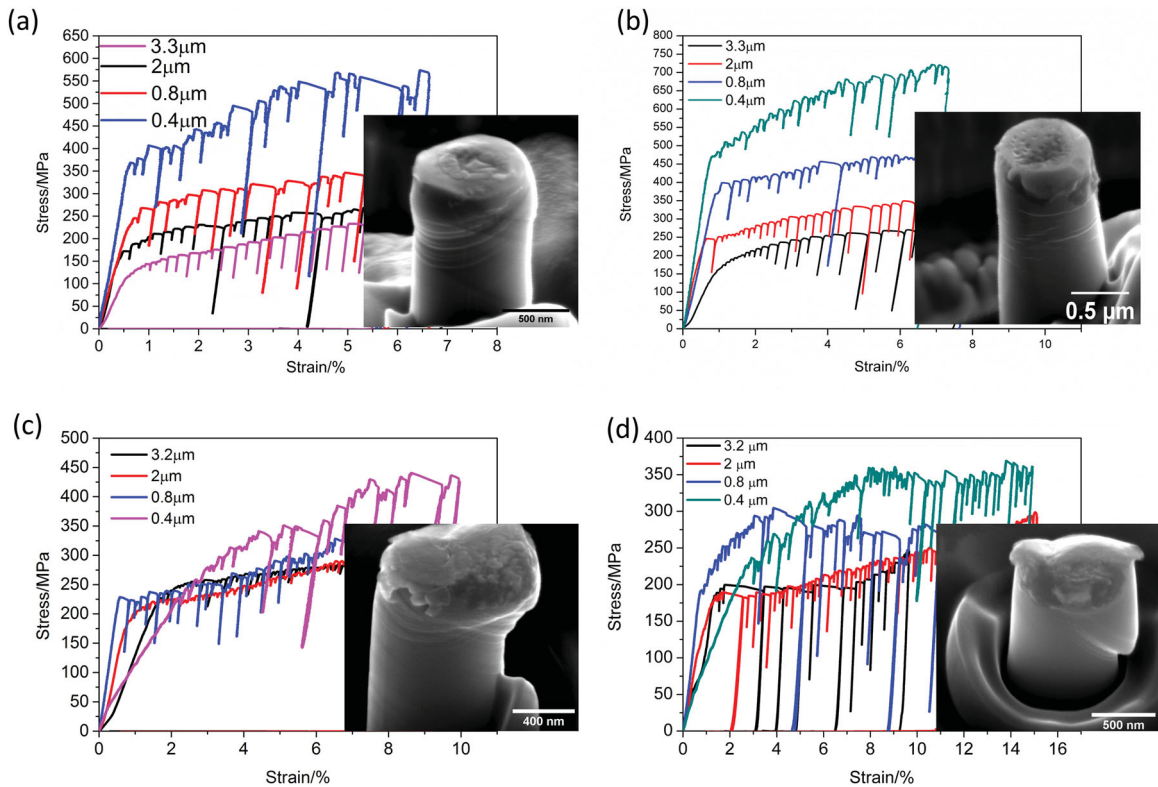


Figure 1. Representative stress-strain curves and the deformed pillar of (a) $[110]$ -orientated ER5 pillar [17], (b) $[111]$ -orientated ER5 pillar [17], (c) $[10\bar{1}]$ -orientated ER4 pillar, (d) $[111]$ -orientated ER4 pillar.

3. Results

3.1. Stress-strain curves

Figure 1(a,b) displays the stress-strain curves for [101]- and [111]-oriented ER5 pillars with diameter ranging from 0.4 to 3.3 μm . Similar to the pure FCC single crystal, the stress-strain curves feature with strain burst and the strength is size-dependent for both orientations, but the deformed morphology is different. The pillar of [101] orientation presents double slip offsets, whereas no obvious slip steps can be seen on the surface of [111]-oriented one. Figure 1(c,d) shows the stress-strain curves for [10-1]- and [111]-oriented ER4 pillars. Although strain burst can be seen in the curves, the yield strength is size-independent. The deformed morphology of [10-1]-oriented ER4 pillar is similar to the ER5 pillar, but the ER4 pillar of orientation [111] has one slip offset that is different from the ER5 pillar. The difference is possibly due to the dislocation density in these two kinds of pillars.

To quantitatively express the size effect on the strength, the experiment results are analysed by the following power law [18]:

$$\sigma = A(D)^m \quad (1)$$

where σ is the yield stress, A is a constant, D is the sample's diameter, and m is the size exponent. The obtained curves are shown in Figure 2(a). For ER4 pillars, their size exponent is very small ($-0.1 \sim -0.03$), indicating that there is no size effect on the strength for the ER4 pillars at nano/microscales. The obtained size exponent for ER5 pillars is about -0.45 , which is smaller than that of the pure FCC Al.

3.2. Dislocation behaviour

Figure 3 shows the dislocation structure of the deformed ER4 and ER5 pillars with $\langle 110 \rangle$ and [111] orientations.

For the ER4 pillars (Figure 3(a-c)), the dislocation structure, featuring with dislocation loops and entanglements, is much more complexed than that of the ER5 pillar (Figure 3(d)). More details about the dislocation structure can be found in the supplementary materials (see Figures S3–S5). The dislocation density is calculated according to the method described by Jennings et al. [24]. The obtained results are shown in Figure S6 in the Supplementary materials. The dislocation density for the ER4 pillar can reach up to 10^{16} m^{-2} , which is two orders higher than that of the ER5 pillar with the same diameter (10^{14} m^{-2}). The difference between these two kinds of pillars can be understood from two facts: the initial dislocation density in ER4 pillars ($1.32 \times 10^{13} \text{ m}^{-2}$), and stronger dragging effect from the solute elements of Mg and Si. For the ER4 pillar, the 2 μm -diameter pillar has higher dislocation density than that of the 0.8 μm -diameter pillar (see Figure S6). For larger samples, the dislocation is more difficult to escape from the sample's free surface due to the impeding force from the solutes and multiplication of dislocation, leading to higher dislocation density.

3.3. Plasticity of pillar with high-dislocation density

To understand the dislocation density on the deformation behaviour of the material in the FZ at nano/microscales, some bigger deformed pillars with orientation $\langle 110 \rangle$ were remade into smaller ones using FIB. The results are shown in Figure 4(a,b). Although the dislocation density is much higher, the stress-strain curves for the pillars still features with strain burst, as illustrated in Figure 4(a,b). However, there is almost no size effect on the strength for both kinds of pillars, as demonstrated in Figure 4(c). Moreover, the increment of yield strength due to the pre-straining is larger in bigger samples, as demonstrated by the 3.3 μm -diameter ER5 pillars

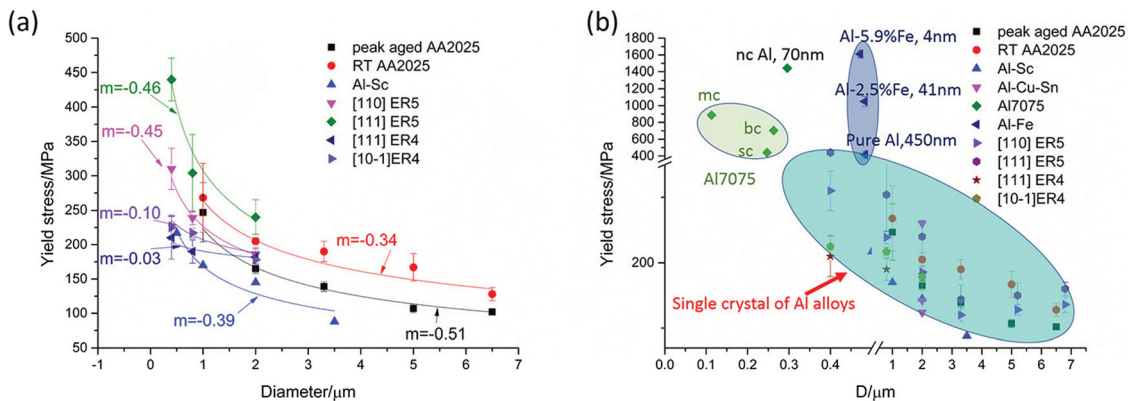


Figure 2. (a) Yield stress vs. diameter for the present study and Refs [17,19–21], (b) effect of alloying and sample's size on the yield stress of Al-based alloys (data from present study and Refs [17,19–23]).

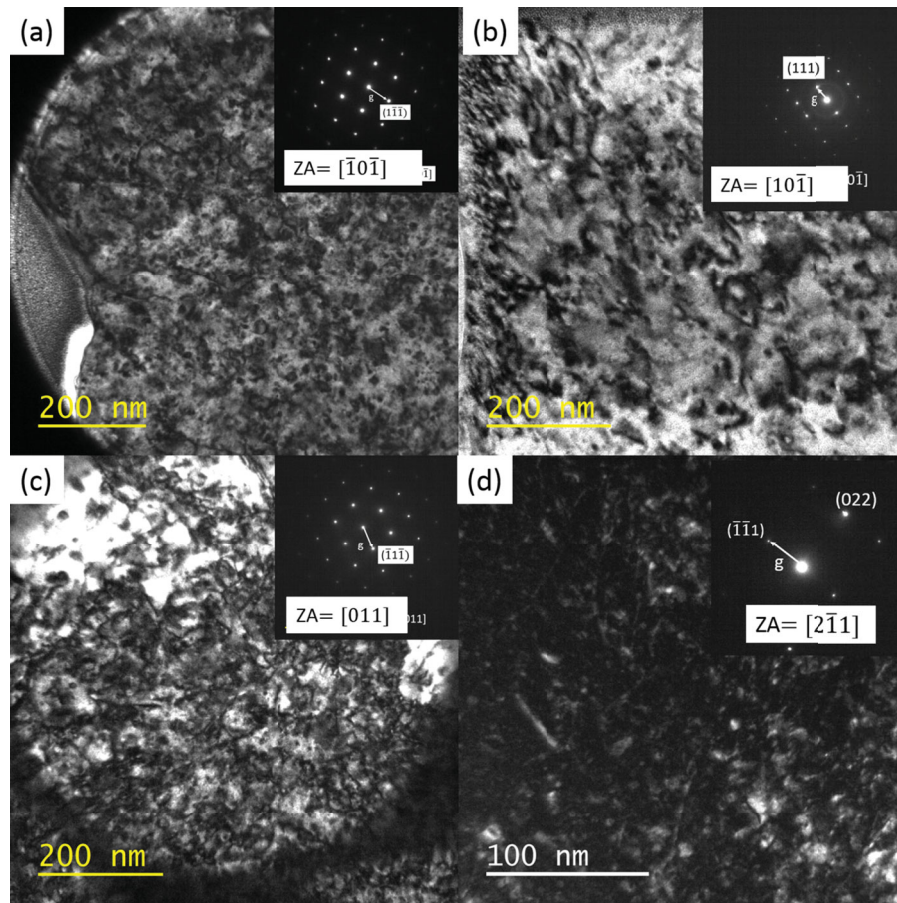


Figure 3. Bright field TEM images of (a) [111]-orientated 800nm-diameter ER4 pillar, (b) [10-1]-orientated 800nm-diameter ER4 pillar, (c) 10-1]-orientated 2000nm-diameter ER4 pillar; (d) dark field TEM image of [111]-orientated 800nm-diameter ER5 pillar.

(Figure 4(c)). In the bigger pillars, a dislocation tends to be trapped in the pillar as it must travel over a longer distance to escape from the free surface. Thus, the dislocation density is higher for bigger pillars and the strength is bigger.

4. Discussion

4.1. Effect of solute elements

For the Al-based alloys (such as AA2025 [19], Al-Sc alloys [20]), as shown in Figure 2(a), the space and size of the internal particle/cluster is much smaller than the pillar's size, nevertheless, the strength of the pillar is still size dependent. It seems that the size-dependent strength is still effective if the particle is not strong enough [25]. For the Al-based alloys, the exponent m in Equation (1) is between -0.34 and -0.51 , derived from the curve shown in Figure 2(a). These results of ER5 pillars are consistent with the researches mentioned above. The lower size effect is due to lower dislocation mobility that is caused by the lattice friction or the dragging effect from these

solute or precipitates. Basically, the material with higher crystal resistance against the dislocation motion exhibits lower size effect [26].

One of the points of interest in studying the mechanical properties at nano/microscales is to finding a way to make the material tougher, or to stabilize the plastic deformation without or reduced strain burst [27,28]. Figure 2(b) summarizes the up-to-date results of the yield stress for Al alloys with different sizes of diameter at nano/micro scales. It can be seen from Figure 2(b) that the yield strength of single crystal of Al alloys is typically in the range from 88 to 400 MPa, regardless of the type of strengthening method (solute, precipitates or both) used. However, if the pillar includes nanoscale strong particles [22] or grain boundaries [23], the yield strength could reach up to 1.5 GPa, as seen in the case of Al7075 and Al-Fe alloys in Figure 2(b). Moreover, the stress-strain curves for the samples with nano-size grains are much smoother than that of the single crystal of Al alloys [20,22,23,28]. This comparison provides a new clue to obtaining higher-strength materials with stable plasticity.

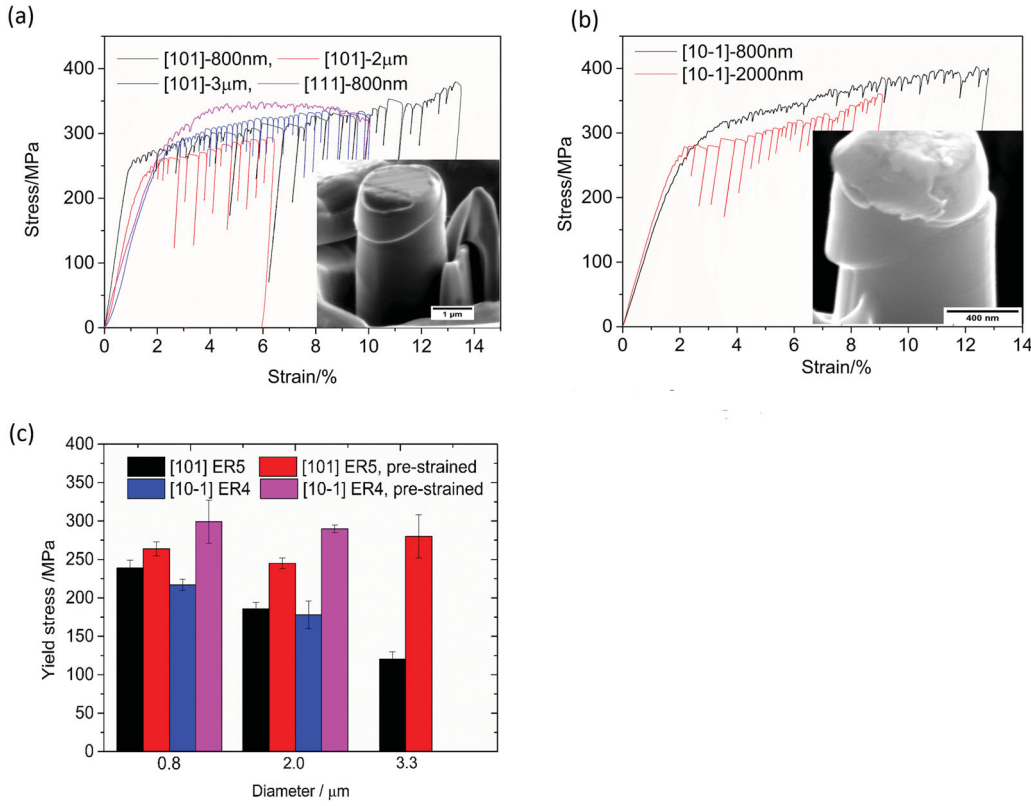


Figure 4. (a) Stress-strain curves for pre-deformed [101]-oriented ER5 pillar, (b) stress-strain curves for pre-deformed [101]-oriented ER5 pillar, (c) yield stress vs. diameter for the ER4 and ER5 pillars.

4.2. Theoretical model

Surprisingly, ER4 pillars show size-independent strength at nano/micro scales, as demonstrated in Figure 2(a). This phenomenon may be brought about by two factors: internal alloying effect and dislocation density. To quantify the effect of the alloying and dislocation density on the strength of Al alloys at microscale, a theoretical model is built, and the possible controlling mechanisms behind is discussed.

The critical resolved shear stress (τ_{CRSS}) of Al alloys can be expressed by [14]:

$$\tau_{CRSS} = \tau_{athermal} + \tau_0 + \tau_{ss} \quad (2)$$

where $\tau_{athermal}$ is the stress from athermal mechanisms associated with the long-range elastic dislocation interactions, τ_0 is the friction stress taken as 10 MPa [16] here, τ_{ss} is the stress from the strengthening mechanisms such as solid solution. $\tau_{athermal}$ can be given as [29]:

$$\tau_{athermal} = \mu \frac{\beta}{D\sqrt{\rho}} + \alpha \mu b \sqrt{\rho} \quad (3)$$

where μ is the shear modulus, D is the diameter of the pillar, b is the magnitude of the Burger's vector, ρ is the dislocation density, β and α are constants which are

determined using the method in [29]. Via experimental data fitting, the constant β for ER5 and ER4 pillars takes 0.003 and 0.004, respectively. the constant α for ER5 and ER4 pillars takes 0.51 and 0.09, respectively. The first term on the right-hand side of Equation (3) represents dislocation source strengthening whereas the second term signifies dislocation forest interactions.

Since only solid solution strengthening is taken into consideration in this study, τ_{ss} can be described as

$$\tau_{ss} = \sum_j K_j C_j^{2/3} \quad (4)$$

where C_j and K_j is the concentration and the scaling factor of the j th alloying element, respectively. Based on the EDS results (Figure S1 in the Supplementary materials) and Ref. [16], τ_{ss} for ER5 and ER4 pillars takes 34 and 54 MPa, respectively. From the quantitative calculation, it can be seen that the dragging effect from solutes in ER4 pillars is greater.

Applying Equation (2), the value of τ_{CRSS} for the pillars with different D and ρ is obtained as shown in Figure 5. Obviously, there is good agreement between the experimental and calculated results for the tested pillars. For the tested pillars, the dislocation-source strengthening decreases with the increase of the size and dislocation density of pillars, as demonstrated in Figure 5. It hints

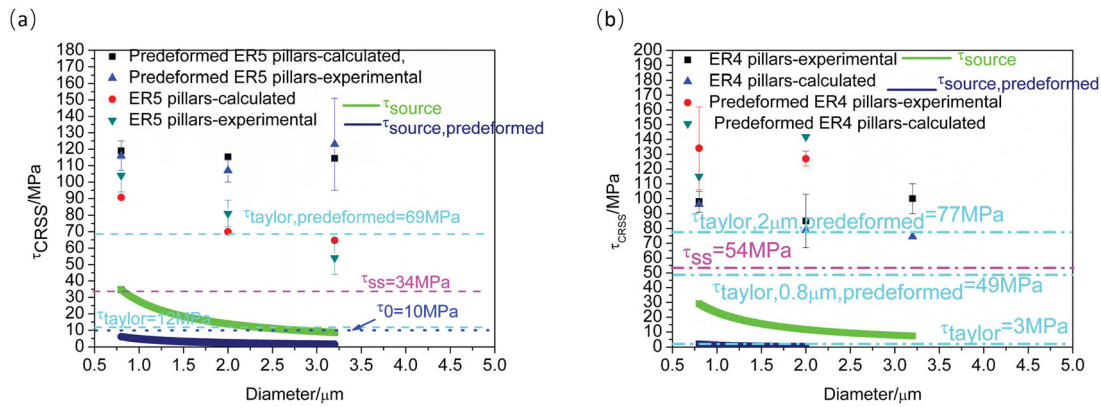


Figure 5. (a) Diameter vs. τ_{CRSS} from experimental results and theoretical model for ER5 and pre-deformed ER5 pillars, (b) Diameter vs. τ_{CRSS} from experimental results and theoretical model for ER4 and pre-deformed ER4 pillars. The contribution to the strength from τ_{source} , τ_{ss} , and τ_{taylor} is illustrated by green, magenta and cyan dashed lines, respectively. For the pre-deformed pillars, the contribution to the strength from τ_{source} is illustrated by navy lines.

that the dislocation density has more influence on the decreasing of the dislocation-source strengthening (see the curves of dislocation-source strengthening in pre-deformed pillars in Figure 5). For 0.8 μm-diameter ER5 pillars (see Figure 5(a)), the strengthening contribution from dislocation source and from solid solution is nearly at the same level, and the strengthening from dislocation interaction is small (12 MPa). However, the contribution from dislocation interaction in pre-deformed ER5 pillars reaches up to 69 MPa, while the dislocation-source strengthening is lower than 10 MPa. This change suggests that the deformation mechanism for the pre-deformed ER5 pillars dominants dislocation interaction.

For ER4 pillars, the highest dislocation-source strengthening is 1.8 times smaller than the solid solution strengthening, suggesting that the pillar's strength may be dominantly controlled by the solid solution strengthening. For the pre-deformed ER4 pillars, the dislocation-source strengthening is almost reduced to 0 MPa due to the high dislocation density. Nevertheless, the strengthening from dislocation interactions is increased up to 77 MPa in 2 μm-diameter pillars. Considering that the solid-solution strengthening is 54 MPa, it is reasonable to infer that both Taylor hardening and solid-solution strengthening are the controlling mechanism.

5. Conclusion

In summary, a serious study was made to investigate the influence of solutes, crystal orientations, sample's size and dislocation density on the nano/microscale deformation of single crystal of solute-strengthening Al alloys. Results showed that the orientation and sample's size still function for ER5 pillars, while the strength of ER4 pillars is size and orientation independent. Increasing the

dislocation density, the ER4 and ER5 pillars exhibit size-independent strength. A theoretical model was built, and the possible controlling mechanisms was discussed.

Acknowledgement

We are appreciated for the support from the group of Prof. Jodie Brady. The authors acknowledge the support from the CAM and ANFF at ANU.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This work is supported by the National Natural Science Foundation of China (No. 11772204). Shaohua Yan would like to thank the financial support from China Scholarship Council for his PhD study.

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